Cryptographic protocols

Lecture 4: Digital signatures

Properties of signatures

- **Signature is authentic**: convinces recipient that the signer deliberately signed the document
- **Signature is unforgeable**: the signer, nobody else, deliberately signed the document
- **Signature is not reusable**: part of the document, can not be moved to a different document
- **Signed document is unalterable**: after signing, document can not be altered
- **Signature can not be repudiated**: The signer can not later claim he/she did not sign the document

Breaking a signature scheme

- Total break: an adversary is either able to compute the private key of the signing or finds an efficient signing algorithm equivalent to a valid signing algorithm.
- Selective forgery: An adversary is able to create a valid signature for a particular message or class of messages chose a priori.
- Existential forgery: An adversary is able to forge a signature for at least one message with little or no control over the message whose signature is obtained.

Attacks against signature schemes

- Key-only attack: adversary only knows the signer's public key
- Message attacks: Adversary is able to examine signatures corresponding either to known or chosen messages:
  - Known-message attack
  - Chosen-message attack (non-adaptive)
  - Adaptive chosen-message attack

Signing with symmetric cryptography

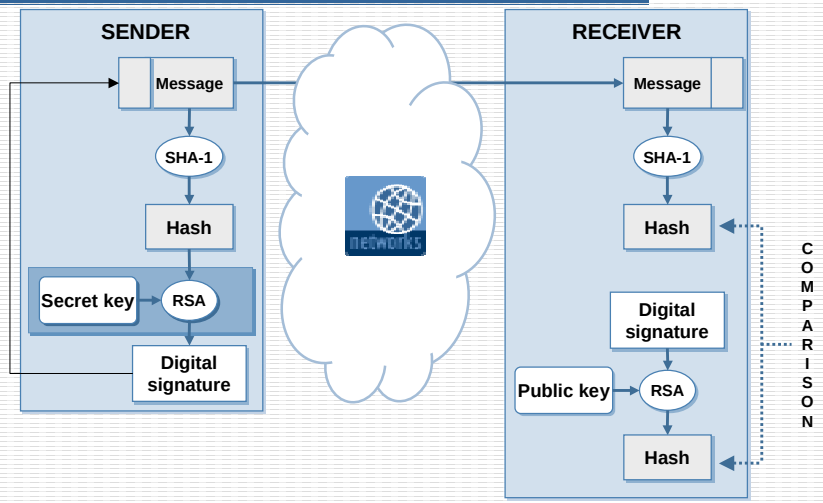
- Requires an arbitrator (T):

1. $A \rightarrow T: \quad E_{K_A}(m)$
2. $B \leftarrow T: \quad E_{K_B}(m, \text{"received from A"})$
3. $B: \quad \text{recovers } m, \text{ assures } m \text{ sent by } A$

Things you already know

- Arbitrators not practical in the real world
- Asymmetric cryptography makes signing feasible without arbitrator
- Usually, only a hash (e.g. 160-bit) signed instead of the entire document
- Timestamps often included in signed messages to prevent recipient cheating
- Non-repudiation requires secure storage of private keys (though is a legal, not technical, property of a signature)

Signing with asymmetric crypto



More formally

- A signs message *m*, B verifies the signature

1. $A: \quad H_m = h(m)$
2. $A: \quad \delta_m = S_A(H_m)$
3. $A \rightarrow B: \quad m, \delta_m$
4. $B: \quad H'_m = h(m)$
5. $B: \quad \delta_m \overset{?}{=} V_A(H'_m)$

Multiple signatures

- A and B want to sign the same document *m*:

1. $A:\ \delta_A = S_A(hash(m))$
2. $B:\ \delta_B = S_B(hash(m))$
3. $A \leftarrow B:\ \delta_B$
4. $A \rightarrow C:\ m, \delta_A, \delta_B$
5. $C:$ verifies δ_A and δ_B

group signatures

- Properties of group signatures:
  - Only members of a group can sign messages
  - The receiver from the signature can verify that it is a valid signature from a group
  - The receiver of the signature can not determine which member of the group is the signer
  - In case of a dispute, the identity of the original signer can be revealed

Group signature protocol

1. T: $K = \{(e_{1,1}, d_{1,1}), (e_{1,2}, d_{1,2}), \ldots, (e_{1,m}, d_{1,m}), \ldots (e_{n,1}, d_{n,1}), \ldots, (e_{n,m}, d_{n,m})\}$
2. $T \rightarrow G_i$: $K_i = \{e_{i,1}, \ldots e_{i,m}\}$ for each group member
3. T: publish public keys (not identities) $K_d = \{d_{1,1}, d_{1,2}, \ldots, d_{n,m}\}$
4. G_i: To sign a document, choose randomly $e' \in K_i$
5. V: Seek corresponding d from the public list, verify signature
6. T: Knows the signer's identity in case of a dispute

Blind signatures

- A sends some information to B for signing, B signs it and returns the signature to A
- From that signature, A can compute B's signature to arbitrary message *m*
- B will know neither *m* nor the signature of *m*
- B can not observe the message signed and later associate it to A
- Applications in e.g. anonymous eCash

Blind signature requirements

- Digital signing mechanism for B, i.e. a signing function $S_B(x)$ for any message x
- Blinding and unblinding functions f and g such that $g(S_B(f(m)))=S_B(m)$. $f(m)$ is called blinded message.
- E.g. blinded RSA
  - let k be a fixed integer gcd(n,k)=1, blinding function is $f(m)=mk^e \bmod n$. unblinding function is $g(m)=k^{-1}m \bmod n$

$$g(S_B(f(m))) = g(S_B(mk^e \bmod n)) = g(m^d k \bmod n) = m^d \bmod n = S_B(m)$$

Chaum's blind signature protocol

- A receives from B the signature of a blinded message, computes from it B's signature for an arbitrary message (selected by A) m. B has no knowledge of m or the signature. s is B's signature on m.

1. $A: \quad k = RND, \quad 1 \le k \le n-1, \gcd(k,n) = 1$
2. $A \rightarrow B: \quad m^* = mk^e \bmod n$
3. $A \leftarrow B: \quad s' = (m^*)^d \bmod n$
4. $A: \quad s = k^{-1}s' \bmod n$

Undeniable signatures

- Signature schemes where co-operation of the signer is required in verification
- For example, when a signer requires that the verifier can not proof the signer's actions without the signer's consent
- For example, Chaum-van Antwerpen undeniable signature scheme

Chaum-van Antwerpen key gen.

- Each entity selects a private key and corresponding public key. A does the following:

1. Select a random prime $p = 2q + 1$ where q is a prime
2. Select a generator α for the subgroup of order q in Z_p^*
   a. Select a random element $\beta \in Z_p^*$, compute $\alpha = \beta^{(P-1)/q} \bmod p$
   b. If $\alpha = 1$ then goto (a.)
3. Select a random integer $a \in \{1,2,\ldots,q-1\}$, compute $y = \alpha^a \bmod p$
4. A's public key is (p, α, y), private key is a

Chaum-van Antwerpen signing

- A signs message *m* so that B can verify the signature with co-operation of A
  1. $A: \quad s = m^a \bmod p$
- Verification protocol:
  1. $B:$ select random integers $x_1, x_2 \in \{1,2,\ldots,q-1\}$
  3. $A \leftarrow B: \quad z = s^{x_1} y^{x_2} \bmod p$
  4. $A \rightarrow B: \quad w = (z)^{a^{-1}} \bmod p$ (where $aa^{-1} \equiv 1 (\bmod p)$
  5. $B: \quad w' = m^{x_1} \alpha^{x_2} \bmod p$ and accept iff $w = w'$

Disavowal

- A can disavow a valid signature by
  1. Refusing to participate in the protocol
  2. Perform the verification protocol incorrectly
  3. Claim that the signature is fake even if it verifies correctly
- A disavowal protocol may be established to guard against (2) and (3)

Chaum-van Antwerpen disavowal protocol

1. B: Obtain A's public key (p, α, y), Select random integers $x_1, x_2 \in \{1,2,\ldots,q-1\}$
2. $A \leftarrow B$: $z = s^{x_1} y^{x_2} \bmod p$
3. $A \rightarrow B$: $w = (z)^{a^{-1}} \bmod p$
5. B: $if\ (w = m^{x_1}\alpha^{x_2} \bmod p)$ Accept, halt
6. B: Select random integers $x'_1, x'_2 \in \{1,2,\ldots,q-1\}$
7. $A \leftarrow B$: $z' = s^{x'_1} y^{x'_2} \bmod p$
8. $A \rightarrow B$: $w' = (z')^{a^{-1}} \bmod p$
9. B: $if\ (w' = m^{x'_1}\alpha^{x'_2} \bmod p)$ Accept, halt
10. B: $c = (w\alpha^{-x_2})^{x'_1} \bmod p, \quad c' = (w'\alpha^{-x'_2})^{x_1}$

- If c=c', B concludes s is forgery, otherwise B concludes s is valid, A attempts to disavow it

Other signature schemes

- Proxy signatures
- Fail-stop signatures
- Designer confirmer signatures
- Many others

Thank you!

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